

Anytime-Valid Conjoint Analysis: An Immigration-Preferences Case Study

Supplementary case study for *Anytime-Valid Inference in Conjoint Experiments* (Molitor, Gosciak, and Lindon, 2026)

July 14, 2026

1 Introduction

The parent paper establishes anytime-valid e-values and confidence sequences (CSs) for the linear-probability (“AMCE”) model of conjoint analysis, and demonstrates their properties through simulation. A companion case study (`cameras_case_study/`) applies the method to a digital-camera conjoint dataset. This note provides a second, independent real-data case study, using the immigration-preferences conjoint experiment of Hainmueller, Hopkins, and Yamamoto (2014) – the paper that originally introduced the AMCE (average marginal component effect) estimator itself, the very estimand the parent paper’s anytime-valid machinery is built to test sequentially. Using their data is thus an especially direct fit for this method: it lets us trace anytime-valid evidence through time on the same dataset, and the same estimand, that motivated the AMCE framework in the first place.

As in the camera case study, the goal here is to show what the method looks like end-to-end on real data – how the confidence sequence and e-value for each attribute effect evolve as respondents accumulate, and how the final anytime-valid interval compares to the conventional fixed- n interval computed from the same fitted model. This dataset is also a useful stress test for the method along a different axis than the camera study: its nine conjoint attributes are all multi-level categorical factors rather than simple binary indicators, producing 41 non-baseline AMCE coefficients instead of ten.

All code, data, and figures for this case study live in `immigration_case_study/` and can be regenerated with `run_all.sh`; see the accompanying `README.md` for exact reproduction steps.

2 Data

We use the `immigrationconjoint` dataset distributed with the R package `cjoint` (Barari, Berwick, Hainmueller, Hopkins, Liu, Strezhnev, and Yamamoto), from the conjoint survey fielded by Hainmueller, Hopkins, and Yamamoto (2014). A nationally representative sample of 1,396 American adults each completed 5 choice tasks; in each task the respondent saw two hypothetical immigrant profiles side by side and chose which one should be admitted to the United States (unlike the camera study, there is no explicit outside/“neither” option – every task forces a choice between the two profiles shown). This gives 13,960 respondent-task-profile rows in total, one per profile shown. The data are distributed already in this long format, so no reshaping is required.

Each profile is described by nine attributes, every one a multi-level categorical factor: **Gender**,

Education, Language Skills, Country of Origin, Job, Job Experience, Job Plans, Reason for Application, and Prior Entry. Two respondent-level columns (`ethnocentrism`) and two pre-computed helper columns (`LangPos`, `PriorPos`) are not used as regressors. We treat each of the 1,396 respondents as a cluster, exactly as in the camera study and the simulations in the main paper, and fit

$$\text{Chosen_Immigrant} \sim \text{Gender} + \text{Education} + \text{Language Skills} + \text{Country of Origin} + \text{Job} + \\ + \text{Job Experience} + \text{Job Plans} + \text{Reason for Application} + \text{Prior Entry}$$

as a linear probability model via `feols()`, clustering standard errors by respondent. This is `cjoint`’s own documented formula for this dataset (see `?cjoint::immigrationconjoint`), used verbatim. Each non-intercept coefficient is an AMCE: the average marginal effect of that attribute level on the probability a profile is chosen, relative to a reference level for that attribute (e.g. India for country of origin, janitor for job, no formal schooling for education) – the same reference levels used throughout Hainmueller, Hopkins, and Yamamoto (2014). Because plain-text CSV storage does not preserve factor level order, `code/02_sequential_analysis.R` explicitly restores these reference levels from the `cjoint` package before fitting any model, rather than letting them default to whatever level sorts first alphabetically.

3 Method

Point estimates and cluster-robust (“CRV1”) standard errors come directly from `feols()`. Anytime-valid p-values and confidence-sequence radii are then computed from those same estimates using the e-value construction of the parent paper, with the reference degrees of freedom and the process index both set to $G - 1$ and G respectively, where G is the number of respondents (clusters) observed so far – not the row count – since clusters, not rows, are the unit that accumulates sequentially here.

The tuning parameter g , which controls the shape of the confidence sequence, is fixed once in advance at the value that minimizes the CS radius at the dataset’s actual final size, $G = 1396$ respondents. This gives $g^* = 169.44$. As in the camera study, g^* is computed once from the planned final sample size and held fixed throughout sequential monitoring below; re-optimizing g as data accumulate would invalidate the anytime-valid guarantee, since g would then depend on how the monitoring unfolds. All significance levels below use $\alpha = 0.05$.

Fitting all nine attributes as multi-level factors produces 41 non-baseline dummy coefficients – far more than the camera study’s 10 binary attributes. `data/final_comparison.csv` and `data/sequential_av_estimates.csv` contain **all** 41 of them; no coefficient is dropped from the underlying results. For the two figures that trace evidence *over time* (Figures 1 and 2), however, faceting all 41 coefficients into small multiples would be unreadable. We instead curate exactly one representative non-baseline level per attribute – nine terms, comparable in number to the camera study’s ten-panel grid – selected by an objective rule: **the non-baseline level with the largest absolute t -statistic in the final full-sample fit, per attribute**. Table 1 shows the resulting selection. The complete 41-coefficient results, curated or not, remain available in `data/final_comparison.csv` and are shown in full in Figure 3.

Table 1: Curated one-per-attribute selection used in Figures 1 and 2: for each attribute, the non-baseline level with the largest $|t\text{-statistic}|$ in the final full-sample fit ($G = 1396$).

Attribute	Selected level	t-statistic
Gender	Male	-3.12
Education	College degree	11.16
Language Skills	Used interpreter	-13.81
Country of Origin	Iraq	-5.80
Job	Doctor	5.75
Job Experience	5+ years	9.92
Job Plans	No plans to look for work	-13.40
Reason for Application	Seek better job	-4.58
Prior Entry	Once w/o authorization	-8.46

4 Results

4.1 Confidence sequences over time

Figure 1 shows each curated attribute’s confidence sequence narrowing steadily as more respondents are observed, while remaining valid to inspect at any point along the way – there is no multiple-looks penalty for having watched, say, the education effect firm up over the first hundred or so respondents before checking in on gender or reason for application much later.

4.2 e-values over time

Figure 2 shows the corresponding e-value trajectories. All nine curated attributes exceed the $1/\alpha$ rejection threshold at some point before the full sample is reached, but the number of respondents needed to first do so varies enormously: from $G = 98$ for Education: College degree to $G = 1287$ for Gender: Male – barely 109 respondents short of the full sample, and only marginally so (its e-value dips back under the threshold at least once more before the study ends). More generally, 3 of the 9 curated trajectories dip back below the threshold at least once after their first crossing; this is expected of e-processes and is harmless under the relevant decision rule, which is to stop and reject the very first time the threshold is crossed, rather than requiring it to stay crossed forever.

4.3 Fixed- n interval vs. confidence sequence, at the final sample size

Figure 3 makes explicit the price of anytime validity at a single, fixed sample size, across the complete set of 41 coefficients: the two methods share the same point estimate in every case (they are computed from the same model fit), but the confidence sequence is visibly wider than the conventional interval for every coefficient. That extra width is exactly what buys the ability to monitor the study continuously, as in Figures 1 and 2, and still report a valid interval or rejection decision at whatever point the analyst chooses to stop – something the conventional interval does not license. The figure also makes clear that not every coefficient’s interval excludes zero at the final sample size: of the 41 non-baseline levels, 25 have a 95% confidence sequence (equivalently, an e-value) that excludes zero (exceeds $1/\alpha$) at $G = 1396$, and 16 do not.

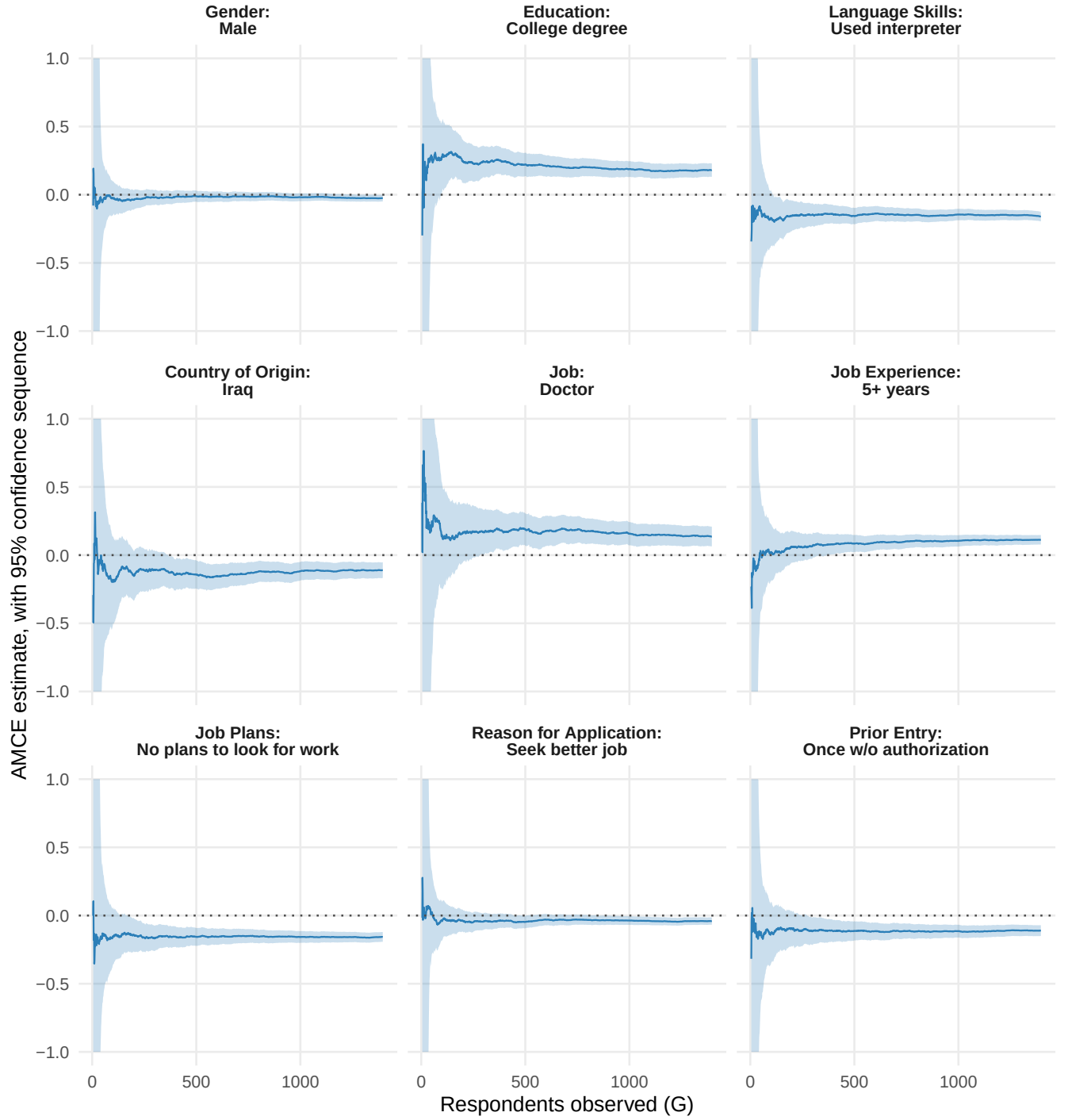


Figure 1: Anytime-valid 95% confidence sequence for the curated one-per-attribute AMCE (Table 1) as respondents accumulate. Shaded regions clipped at ± 1 ; very early on (roughly $G < 63$) the confidence sequence is either unbounded ($G \leq 33$, where $\nu = G - 1$ is too small to be informative) or wider than the plotted range. This is expected: a confidence sequence must sometimes be uninformative to guarantee valid coverage at every sample size, not just at a single pre-planned n .

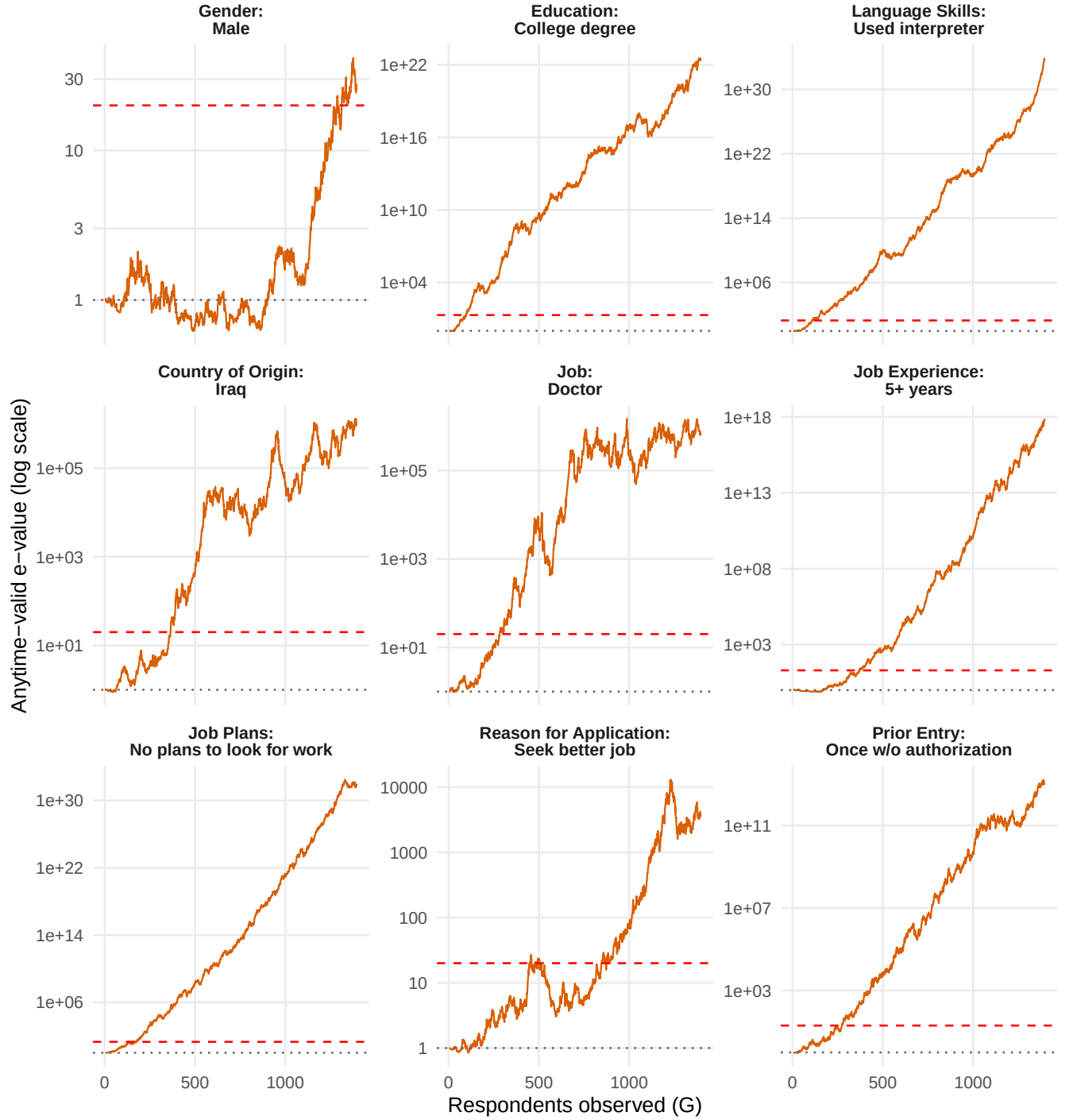
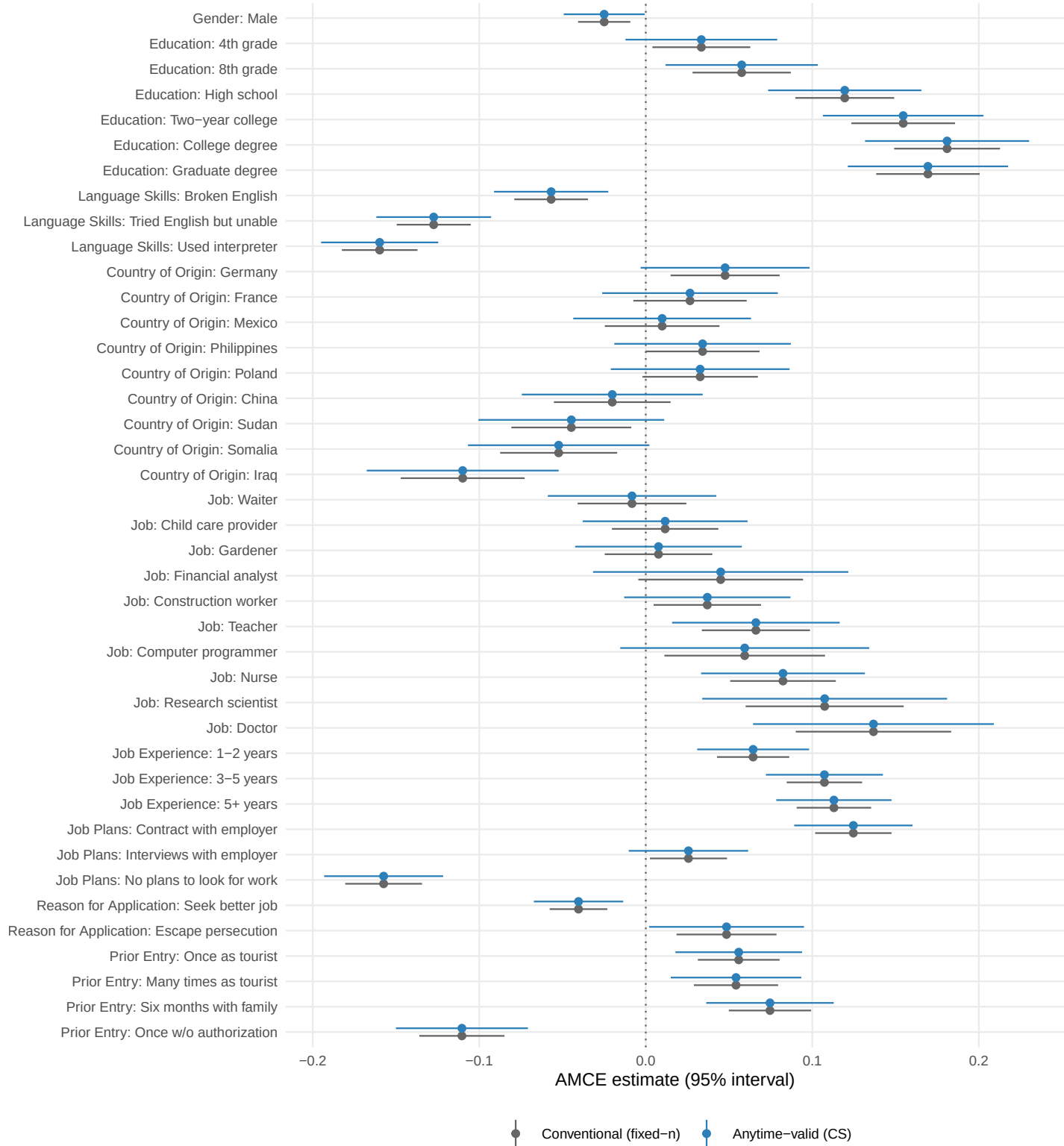


Figure 2: Anytime-valid e-value for the curated one-per-attribute AMCE (Table 1) as respondents accumulate (log scale). The dashed red line is $1/\alpha$, the threshold an e-value must cross to reject the null of no effect at level α ; the dotted grey line is $e = 1$ (neutral evidence).



5 Discussion

Unlike the camera study, where every attribute’s effect was eventually detectable, here the picture is more mixed – itself a realistic feature of this dataset, not an artifact of the anytime-valid method: of the 41 non-baseline AMCEs, 25 have anytime-valid evidence against the null of no effect by the end of data collection, while 13 never cross the rejection threshold at any point during the study, mostly specific **Job** categories (relative to janitor) and **Country of Origin** levels (relative to India) whose effects on admission preferences are small and not detectable at this sample size. A further 3 – Country of Origin: Germany, Country of Origin: Somalia, Country of Origin: Sudan – cross the threshold at some point but fall back below it by the final sample; notably, all three are **Country of Origin** levels. This is exactly the scenario the “stop at first crossing” decision rule is designed for: an analyst monitoring any of these three attributes anytime-validly would have correctly rejected the null when the e-value first crossed the threshold, even though a fixed- n analysis run only once, at the final sample, would have missed that evidence entirely.

Among the nine curated, one-per-attribute AMCEs, the point estimates agree with the qualitative pattern in Hainmueller, Hopkins, and Yamamoto (2014): having a college degree (positive), more job experience (positive), and a high-status job like doctor (positive) all raise the chance a profile is chosen, while needing an interpreter to speak English (negative), having no plans to look for work (negative), a prior unauthorized entry (negative), coming from Iraq rather than India (negative), applying to seek a better job rather than reunite with family (negative), and being male rather than female (negative) all lower it. As in the camera study, the case study illustrates the two costs and benefits of the anytime-valid approach side by side: wider intervals at any fixed sample size, in exchange for validity at every sample size – including whichever one an analyst actually decides to stop at, and including attributes, like several of the **Country of Origin** levels here, whose evidence appears only transiently along the way.

6 References

- Hainmueller, J., Hopkins, D., and Yamamoto, T. (2014). Causal Inference in Conjoint Analysis: Understanding Multi-Dimensional Choices via Stated Preference Experiments. *Political Analysis*, 22(1), 1-30.
- Barari, S., Berwick, E., Hainmueller, J., Hopkins, D., Liu, S., Strezhnev, A., and Yamamoto, T. (2026). cjoint: AMCE Estimator for Conjoint Experiments. R package version 2.1.3.
- Lindon, M., Ham, D., Tingley, M., and Bojinov, I. (2026). Anytime-valid linear models and regression adjusted causal inference in randomized experiments. *Journal of the American Statistical Association*. arXiv:2210.08589.
- Molitor, D., Gosciak, J., and Lindon, M. (2026). Anytime-Valid Inference in Conjoint Experiments.